

Factors, Multiples & Primes

HCF · LCM · Prime factorisation · Factor trees

Key Vocabulary — Mathematical ESL Guide

Prime Numbers — The Sieve of Eratosthenes

How the sieve works: (1) List all numbers from 2 to 50. (2) Cross out multiples of 2 (keeping 2). (3) Cross out multiples of 3 (keeping 3). (4) Continue for 5, 7... (5) All remaining numbers are PRIME.

Factor Trees — Prime Factorisation

Method: Split the number into two factors. Keep splitting each factor until every branch ends in a prime. Circle the primes — they are your answer.

HCF and LCM — Using Prime Factorisation

Knowledge Check — Questions & Model Answers

- Prime factorisation of 90? → $90 = 2 \times 45 = 2 \times 9 \times 5 = 2 \times 3^2 \times 5$. Always use index notation. Check: $2 \times 9 \times 5 = 90$ ✓
- HCF of 48 and 36? → $48 = 2^4 \times 3$ | $36 = 2^2 \times 3^2$. Shared: $2^2 \times 3 = 12$. HCF = 12. Check: $48 \div 12 = 4$ ✓ $36 \div 12 = 3$ ✓
- Two lights flash every 8 s and 12 s. When do they flash together? → LCM(8,12). $8 = 2^3$, $12 = 2^2 \times 3$. LCM = $2^3 \times 3 = 24$ seconds.
- Is 91 prime? → NO. $91 = 7 \times 13$. Classic trick question — always test divisibility by small primes before declaring a number prime.
- LCM of 60 and 84? → $60 = 2^2 \times 3 \times 5$ | $84 = 2^2 \times 3 \times 7$. LCM = $2^2 \times 3 \times 5 \times 7 = 420$. HCF = $2^2 \times 3 = 12$.

